Sample Questions for CO1 – AT1 – Data Interpretation

1. Employee Salary Analysis

A CSV file (employees.csv) contains employee details.

Task:

1. Read the CSV file.
2. Display employees earning more than ₹50,000.
3. Calculate average salary department-wise.

Code:

```
import pandas as pd


df = pd.read_csv("../Csv/Employee.csv")


print("Employees earning more than ₹50,000")
print(df[df["Salary"] > 50000])


print("\nAverage Salary Department-wise")
print(df.groupby("Department")["Salary"].mean())
```

Output:

```
6          107         Arjun         IT   68000
7          108         Meera         HR   58000

Average Salary Department-wise
Department
Finance    51500.000000
HR         51666.666667
IT         67666.666667
Name: Salary, dtype: float64
```

2. Missing Data Handling

Given a CSV file containing student marks with missing values.

Task

1. Identify missing values.
2. Replace missing marks with the column average.
3. Display the cleaned dataset.

Code:

```
import pandas as pd


df = pd.read_csv("students.csv")


print("Missing Values")

print(df.isnull())


df.fillna(df.mean(numeric_only=True), inplace=True)


print("\nCleaned Dataset")

print(df)
```

Output:

```
4    False  False  False    False    False

Cleaned Dataset
   Roll_No   Name  Maths  Science  English
0        1  Rahul  85.00     90.0    88.00
1        2  Priya  78.00     86.0    92.00
2        3   Amit  90.00     85.0    89.75
3        4  Sneha  82.25     89.0    95.00
4        5  Karan  76.00     80.0    84.00
```

3. Duplicate Record Removal

A customer dataset contains duplicate rows.

Task:

1. Read the dataset.
2. Find duplicate records.
3. Remove duplicates.
4. Display the updated dataset.

Code:

```
import pandas as pd


df = pd.read_csv("../Csv/customer.csv")


print("Duplicate Records")

print(df[df.duplicated()])


df = df.drop_duplicates()


print("\nUpdated Dataset")

print(df)
```

Output:

```
5          101  Rahul     Chennai

Updated Dataset
   Customer_ID   Name        City
0          101  Rahul     Chennai
1          102  Priya  Coimbatore
2          103   Amit     Madurai
4          104  Sneha       Salem
6          105  Karan      Trichy
```

4. Data Transformation

A product dataset stores prices in USD.

Task

1. Convert prices into INR.
2. Add a new column called "Price_INR

Code:

```
import pandas as pd


df = pd.read_csv("../Csv/product.csv")


exchange_rate = 83


df["Price_INR"] = df["Price_USD"] * exchange_rate


print(df)
```

Output:

```
   Product_ID Product_Name  Price_USD  Price_INR
0         101       Laptop        800      66400
1         102        Mouse         20       1660
2         103     Keyboard         35       2905
3         104      Monitor        250      20750
4         105   Headphones         50       4150
```

5. Data Aggregation

Given monthly sales data.

Task

1. Calculate total sales.
2. Average sales.
3. Maximum sales.
4. Minimum sales.

Code:

```
import pandas as pd


df = pd.read_csv("monthly_sales.csv")


print("Total Sales:", df["Sales"].sum())

print("Average Sales:", df["Sales"].mean())

print("Maximum Sales:", df["Sales"].max())

print("Minimum Sales:", df["Sales"].min())
```

Output:

```
Total Sales: 190000
Average Sales: 31666.666666666668
Maximum Sales: 40000
Minimum Sales: 25000
```

6. Grouping Data

Given employee information.

Task

1. Group employees by department and calculate:
2. Total salary
3. Average salary
4. Employee count

Code:

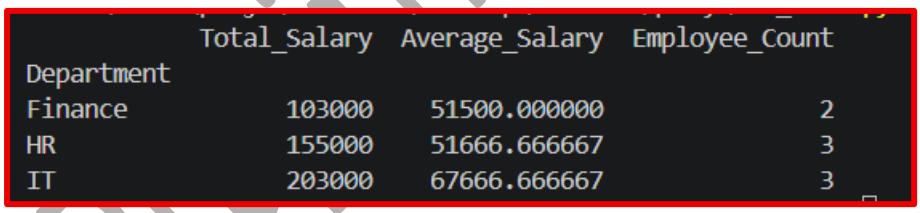
```
import pandas as pd


df = pd.read_csv("employees.csv")


result = df.groupby("Department")["Salary"].agg(
    Total_Salary="sum",
    Average_Salary="mean",
    Employee_Count="count"
)


print(result)
```

Output:



Department	Total_Salary	Average_Salary	Employee_Count
Finance	103000	51500.000000	2
HR	155000	51666.666667	3
IT	203000	67666.666667	3

7. XML Data Analysis

Given library.xml

Task

Display:

1. Total books
2. Books written after 2020
3. Books costing more than ₹500

Code:

```
import xml.etree.ElementTree as ET


tree = ET.parse("library.xml")

root = tree.getroot()


print("Total Books:", len(root.findall("book")))


print("\nBooks written after 2020:")

for book in root.findall("book"):

    if int(book.find("year").text) > 2020:

        print(book.find("title").text)


print("\nBooks costing more than ₹500:")

for book in root.findall("book"):

    if float(book.find("price").text) > 500:

        print(book.find("title").text)
```

Output:

```
Books written after 2020:
Data Science
Machine Learning
Deep Learning

Books costing more than ₹500:
Data Science
Machine Learning
Artificial Intelligence
Deep Learning
```

8. Data Interpretation using Python

Given marks of students.

Task

1. Store marks in a list.
2. Find:
3. Highest mark
4. Lowest mark
5. Average mark
6. Number of students passed.

Code:

```
marks = [78, 65, 90, 45, 35, 88, 55, 72]


print("Marks:", marks)

print("Highest Mark:", max(marks))

print("Lowest Mark:", min(marks))

print("Average Mark:", sum(marks) / len(marks))


passed = 0

for mark in marks:

    if mark >= 40:

        passed += 1


print("Number of Students Passed:", passed)
```

Output:

```
Marks: [78, 65, 90, 45, 35, 88, 55, 72]
Highest Mark: 90
Lowest Mark: 35
Average Mark: 66.0
Number of Students Passed: 7
```

9. Dictionary-Based Analysis

Store employee information using dictionaries.

Task

Display:

1. Highest salary employee
2. Lowest salary employee
3. Average salary

Code:

```
employees = {
   "Rahul": 45000,
   "Priya": 60000,
   "Amit": 55000,
   "Sneha": 75000,
   "Karan": 52000
}

highest = max(employees, key=employees.get)
lowest = min(employees, key=employees.get)
average = sum(employees.values()) / len(employees)

print("Highest Salary Employee:", highest, "-", employees[highest])
print("Lowest Salary Employee:", lowest, "-", employees[lowest])
print("Average Salary:", average)
```

Output:

```
Highest Salary Employee: Sneha - 75000
Lowest Salary Employee: Rahul - 45000
Average Salary: 57400.0
```